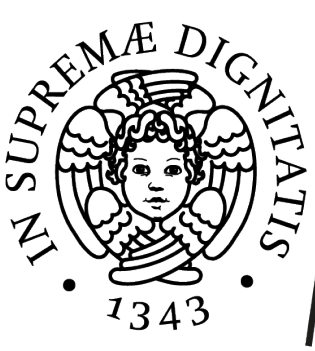

UNIVERSITÀ DI PISA

Department of Philology, Literature, and Linguistics

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GRADUATION THESIS

MAI(A) the Sound Be With You

MAIA-AV: A Ground-Up Audio-Visual Benchmark for Multimodal AI Assessment

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Introduction

The rapid evolution of Visiual Language Models (VLMs) raises the need to determine whether their strong performance reflects genuine multimodal understanding or the exploitation of superficial correlations and linguistic biases. In this contex *MAIA: Multimodal AI Assessment* by Testa et al.(2025)[20, 19] was introduced as a native Italian benchmark for evaluating video-based reasoning in Vision-Language Models. Testa et al. investigate the extent to which VLMs coherently integrate visual and linguistic information. More specifically, it assesses whether model predictions are grounded in visual evidence rather than primarily driven by distributional regularities learned by the language component. This distinction is crucial because a model may answer a question correctly while failing to recognise the same information consistently when it is presented in a discriminative format. In such cases, the prediction may result from linguistic plausibility rather than genuine visual understanding. To address this issue, MAIA employs two aligned tasks: Visual Statement Verification and Open-Ended Visual Question Answering. Through these tasks, Testa et al. evaluate not only models accuracy, but also the robustness and consistency of models behaviour across different evaluation formats. The benchmark further defines twelve fine-grained reasoning categories to identify which competencies are used by the models and in which areas linguistic shortcuts, unimodal collapse, or hallucinations emerge. A limitation of the original framework is that the dataset was designed and evaluated primarily on the basis of visual information. Real-world videos, however, are inherently audiovisual: relevant meaning may also be conveyed through dialogue, environmental sounds, music, and other acoustic events. Recent multimodal and omnimodal architectures can process visual, auditory, and textual inputs, but the availability of multiple modalities does not guarantee that they will be integrated effectively. Models may over-rely on a single channel or be negatively affected by redundant or weakly relevant information. Starting from this problem, the study Deodati,Kaif et al.(2026): "*Lost in Transcription: Investigating the Role of Audio Information for Video-based Visual Statement Verification*"[4] extends the original MAIA research framework to a more complex

analysis of the interaction between visual, linguistic, and auditory information, while preserving its competence-oriented perspective. The study considers two complementary representations of audio: raw signal and automatic transcriptions provided as textual input. By evaluating these representations both independently and in combination with video information, the analysis investigates whether audio contributes meaningfully to task performance and whether one representation has a stronger influence on model predictions. This leads to two research questions:

RQ1: *To what extent does raw audio affect video-based Visual Statement Verification when it is jointly processed with visual and textual information?*

RQ2: *How effectively do multimodal models integrate visual, textual, and auditory information when performing video-based reasoning and understanding tasks?*

The experiments show that raw audio has a limited and strongly context-dependent effect. It is most useful when visual information is unavailable, whereas, when added to an already informative visual input, it may provide little benefit or introduce noise. The results also show that transcriptions can produce greater interference than raw audio, suggesting that multimodal architectures do not necessarily achieve more effective cross-modal understanding simply because additional modalities are available. These findings must nevertheless be interpreted in relation to the nature of the dataset. The one hundred original MAIA videos were selected and annotated according to their visual content, and the questions were formulated without considering the audio track. Consequently, the benchmark does not systematically require auditory information to solve its items. My thesis project begins from this limitation. The following chapters investigate whether models can integrate multiple input modalities more coherently when the dataset is specifically designed so that auditory information is at least partially relevant. The central hypothesis is that, under these conditions, models may use complementary signals across modalities to reconstruct missing information and develop a more genuinely multimodal understanding of video content, producing answers grounded in the available evidence rather than in linguistic plausibility or chance.

1. Literature review

The rapid evolution of Visual Language Models (VLMs) has progressively expanded the ability of neural systems to jointly process perceptual and linguistic information. This line of research builds on the hypothesis that linguistic meaning cannot be constructed exclusively from distributional regularities learned from textual corpora, but must also be grounded in perceptual experience [2]. Within this context, Visual Question Answering (VQA) emerged as one of the first systematic paradigms for assessing whether a model can answer questions about the content of an image [1]. Subsequent benchmarks, such as GQA by Hudson and Manning (2019), extended this setting by introducing compositional questions involving entities, attributes, and visual relations [8]. However, high task accuracy alone does not provide sufficient evidence of multimodal comprehension. Several studies have shown that models can exploit statistical regularities in answers, questions, or dataset distributions, producing correct predictions without necessarily *understanding* the associated visual content. Evaluation has therefore progressively moved beyond task performance towards the assessment of grounding, understood as the ability to associate a prediction with the relevant perceptual evidence. One widely adopted strategy for evaluating grounding relies on minimally contrastive pairs, in which a correct instance is paired with a foil obtained by modifying a semantically relevant element. *FOIL-it!*, for example, evaluates whether a model can detect localized inconsistencies between an image and its linguistic description [17]. *Winoground*, instead, presents pairs of images and descriptions containing the same lexical elements but arranged according to different compositional relations, showing that recognizing individual entities does not necessarily entail understanding the relational structure connecting them [21]. The transition from images to videos introduces additional challenges. Video understanding requires integrating information distributed over time, recognizing actions and state changes, ordering events, and establishing temporal and causal relations. *Action Genome Question Answering (AGQA)* addresses these requirements through compositional questions concerning objects, actions, and spatio-temporal relations [7]. Similarly, the *Multi-modal*

Video Understanding Benchmark (MVBench) evaluates a broader range of competencies, including action perception, movement, temporal ordering, and dynamic reasoning [12]. Nevertheless, video benchmarks remain vulnerable to unimodal shortcuts and to strategies that rely predominantly on the most informative frame. Kesen et al. (2024) address this limitation with *Video Language Model Assessment (VILMA)*, a zero-shot benchmark based on video–text contrastive pairs [9]. For each complex capability, VILMA introduces a proficiency test designed to verify the preliminary perceptual ability required to solve the corresponding task. The results show that, once these prerequisite conditions are considered, part of the apparently correct performance can be attributed to accidental or spurious strategies. In particular, the examined VLMs do not display a systematic advantage over statistical image-based models on tasks requiring strict temporal grounding. Within this line of research, MAIA introduced a natively Italian benchmark designed to provide a fine-grained evaluation of video-linguistic reasoning [20]. The resource covers twelve reasoning categories related to linguistic phenomena, including causal, counterfactual, implicit, spatial, temporal, sentimental, planning, uncertainty, and out-of-scope reasoning. Its main contribution lies in aligning two tasks built on the same videos and questions: *Visual Statement Verification (VSV)*, where the model selects the correct statement from a minimally contrastive pair, and *Open-Ended Visual Question Answering (OEVQA)*, where the model freely generates the answer. This setting makes it possible to assess not only accuracy but also coherence between discriminative comprehension and linguistic generation. A model may correctly answer a multiple-choice question while failing to reproduce the same information in an open-ended setting, or may behave inconsistently across equivalent linguistic formulations. To capture this phenomenon, MAIA introduces the *Aggregate Accuracy* metric, which rewards only those instances in which the model correctly answers all true–false pairs associated with the same question and simultaneously generates the correct response in the corresponding open-ended task. The reduction in Aggregate Accuracy relative to conventional single-instance accuracy highlights the fragility of apparently strong performance and the need to jointly evaluate robustness, consistency, and grounding. Although VLMs have traditionally focused on text–vision interactions, real videos are intrinsically audiovisual. Their interpretation may depend on

music, prosody, background noise, and other acoustic signals that cannot be reconstructed from visual frames alone. Consequently, an increasing body of research has extended multimodal language model architectures to support joint processing of text, images, video, and audio. *ImageBind*, for instance, introduced a shared representation space spanning images, text, audio, depth, thermal information, and inertial movement [6]. Other multimodal language models (MLMs) rely on modality-specific encoders combined with projection or fusion modules that align heterogeneous representations with the linguistic space. *Video-SALMONN: Speech-Enhanced Audio-Visual Large Language Models* introduces a multi-resolution Q-former[1] to align audio signals with synchronized video [18], whereas *VideoLLaMA2* combines spatio-temporal modeling and acoustic processing within a generative multimodal architecture [3]. More recent omnimodal architectures process multiple modalities within a unified framework. *Qwen2.5-Omni* adopts a thinker–talker architecture that accepts text, image, video, and audio inputs and produces textual or audio responses [13]. The subsequent *Qwen3.5-Omni* architecture integrates visual and audio encoders into a common backbone and introduces explicit temporal references to improve audio–video alignment [14]. These developments demonstrate that heterogeneous signals can be processed within a single architecture. However, the availability of a modality does not imply that it is used functionally or complementarily. This distinction between **multimodal availability** and **multimodal integration** is therefore essential. The former indicates that a system can accept multiple input sources, the latter requires the model to select, coordinate, and combine relevant information across modalities. A system may consequently be multimodal from an architectural perspective while relying predominantly on unimodal information during prediction. This issue is evident in *Audio-Visual Question Answering* benchmarks such as MUSIC-AVQA, which evaluates spatio-temporal relations in musical performance videos and extends the analysis to objects, entities, and everyday activities [11]. Subsequent studies reveal a strong visual bias, showing that many questions can be answered using the video stream alone, without exploiting the

[1] A multi-resolution Q-former, such as a Multi-Resolution Causal Q-Former, processes continuous audio, video, or image information at different temporal or spatial scales through distinct query banks, balancing fine-grained signals with broader contextual information.

audio channel. Under these conditions, similar audio-only and video-only performance does not demonstrate genuine integration, but rather suggests a form of unimodal collapse in which one modality dominates the decision process. To address this problem, Radevski, Gorjan, et al. (2026) introduced *Dave*, a benchmark in which each question jointly requires visual and acoustic information, making either modality insufficient in isolation [15]. The benchmark distinguishes among action recognition, sound classification, temporal ordering, and multimodal synchronization. This decomposition makes it possible to determine whether an error arises from the failure to perceive an event or from the inability to establish temporal relations between sounds and visual actions. The results show that models may achieve strong performance on individual components while encountering greater difficulty in audio–visual synchronization, indicating that cross-modal integration remains a central limitation. The increasing attention to audio has also led to benchmarks specifically designed to assess acoustic intelligence. *MMAU-Pro* evaluates 49 competencies distributed across speech, environmental sounds, music, and their combinations [10]. These tasks include long-audio comprehension, multi-track reasoning, spatial localization, prosody interpretation, and the integration of dialogue, music, and background noise. The persistently limited performance of current models indicates that acoustic processing cannot be reduced to dialogue recognition alone, since audio also conveys spatial, temporal, emotional, and contextual information that must be evaluated independently. A complementary perspective is provided by *SONIC-01*, a benchmark for omnimodal models based on audio–visual interactions drawn from real conversational and everyday contexts [16]. SONIC-01 combines open-ended summarization, multiple-choice questions, and temporal event localization, while also emphasizing that replacing raw audio with transcription removes paralinguistic information related to tone and communicative intention. Its results identify temporal localization as one of the most challenging tasks, confirming that the integration of verbal content, acoustic signals, and visual sequences remains an open problem. The comparison between native audio and transcription is also central to my recent work, *Lost in Translation: Investigating the Role of Audio Information for Video-based Visual Statement Verification* [4]. The study extends the MAIA framework by introducing audio into the VSV task and comparing two representations of the acoustic modality: raw

audio directly processed by *Qwen2.5-Omni-7B* and automatic transcriptions supplied as additional textual input to *Qwen2.5-VL-7B*. To characterize the audio content, the videos are divided into four classes:

Music: Instances characterized by repetitive vocalizations, onomatopoeic patterns, strongly repetitive tokens, or explicit lexical cues associated with non-speech audio.

Dialogue: Videos whose transcriptions display sufficiently coherent linguistic content, acceptable log-probability values, low degeneracy, and limited evidence of no-speech segments.

Silence/Noise: Videos with empty or highly unreliable transcriptions, particularly when associated with low log-probability values and relatively high no-speech probabilities.

Unknown: Residual cases that appear speech-related but do not satisfy the dialogue criteria with sufficient confidence, thereby preserving potentially informative content while explicitly marking its uncertainty.

Class labels and speech transcriptions were obtained by combining the outputs of *Whisper-1* and *GPT-4o-transcribe*, together with an assessment of output plausibility. This classification makes it possible to distinguish results according to the informativeness of the available audio content. The results reveal a marked dominance of the visual channel. Audio provides only a limited benefit, mainly when no visual information is available, once rich visual input is provided, its contribution becomes negligible and may even introduce noise or errors, with limited robustness when a single black frame is added. Under visually rich conditions, the difference between with-audio and without-audio settings is therefore minimal, suggesting that audio functions more as a surrogate channel than as information fully integrated with vision. Audio transcriptions exhibit a different behavior. When added to already rich visual input, they reduce accuracy relative to the visual-only condition. This result shows that transcription is not a neutral transformation: once projected into the same representational space as the questions and answer options, it can compete with them, increasing the influence of textual regularities and introducing redundant or

irrelevant information. *Lost in Translation* therefore shows that different representations of the same acoustic content can affect the model's decision process differently. Raw audio can be partially ignored when it is not useful, whereas transcription directly interacts with the linguistic component of the model. The introduction of an additional modality consequently does not guarantee an improvement and may instead compromise predictions that were already correct on the basis of reliable visual evidence. These findings must be interpreted in light of the original MAIA design. The benchmark was created to investigate visual content, and annotators were explicitly instructed not to consider audio information. Consequently, its questions and answers were not designed to exploit acoustic evidence. The limited contribution of audio may therefore reflect both the models' modality-integration mechanisms and the limited relevance of audio within the original dataset. Recent literature further emphasizes the importance of identifying the distinct sources of multimodal errors. *MIRAGE* disentangles hallucinations caused by perceptual failures from those arising from incorrect reasoning over correctly represented information [5]. Although primarily focused on visual reasoning, its methodological principle extends naturally to audio–visual settings, where an incorrect response may derive from failed sound recognition, inaccurate temporal synchronization, selection of the wrong modality, or erroneous inference over otherwise correctly represented evidence. Complementary attribution approaches seek to identify which parts of the input contribute to the generation of an answer. *OmniTrace* proposes an attribution framework for omnimodal autoregressive generation that links generated tokens to textual segments, visual regions, and audio/video intervals by aggregating attention signals into cross-modal explanations associated with linguistically interpretable units. This type of analysis is particularly relevant for determining whether a model actually exploits audio information or whether its prediction is driven primarily by text and vision. The current state of the art therefore reveals a clear tension between the rapid development of omnimodal architectures and their effective ability to integrate heterogeneous modalities. Contemporary models can receive multiple input types within the same context, yet diagnostic benchmarks provide no guarantee that these sources are used complementarily. Performance may instead depend on a dominant modality, linguistic shortcuts, dataset correlations, or additional signals that in-

troduce noise. A central unresolved limitation is the construction of benchmarks in which audio and visual information are explicitly supervised. Assessing genuine audio–visual comprehension requires items in which audio contributes relevant information and the correct answer cannot be determined from a single modality alone. Such a design would make it possible to distinguish complementarity, dominance, substitution, and interference, while evaluating whether the model selects the appropriate evidence and combines it coherently across modalities. The present study follows this direction. Building on the competence-oriented MAIA framework and the recent findings of *Lost in Translation*, it investigates model behavior on a dataset specifically designed to require a genuine contribution from audio in order to answer the questions correctly. The central hypothesis is that, when audio and video provide partial and complementary evidence, multimodal comprehension depends on the model's ability to reconstruct the missing information through cross-modal integration, thereby avoiding both unimodal collapse and linguistic shortcuts based on plausibility.

2. Data Collection

2.1 From MAIA to MAIA-AV: Dataset Expansion and Curation

Video Selection

The dataset adopted in MAIA-AV builds upon the guidelines of Multimodal AI Assessment (MAIA), comprising 100 short videos, approximately 30 seconds each, selected from YouTube Italia to represent everyday scenarios of Italian culture. Content domains include sports, culinary arts, urban life, and cultural activities, with a strong focus on human presence and highly informative scenes. In the original formulation, video selection and annotation were oriented strictly toward visual comprehension, completely omitting audio content. Building upon this initial set, we collected an additional 100 videos, each uniquely identified from `video001` to `video100`. These identifiers map each video to its corresponding audio stream and text transcriptions. Each video was precisely trimmed to extract targeted segments, creating a controlled corpus of clips. Particular attention was paid to the alignment between dialogue content and the elements appearing in the scene. This targeted selection of relevant dialogue enables a precise investigation of multimodal interactions. By prioritizing scenarios where spoken dialogue is semantically grounded in the visual content, we ensure that the audio signal plays a genuinely informative role in the multimodal instance. This distinction represents the core departure from the original dataset: MAIA-AV is a benchmark engineered to combine and compare distributed evidence across visual, auditory, and textual modalities.

Implementation Details The video acquisition pipeline relies on `yt_dlp` to retrieve targeted temporal segments from selected YouTube videos. Rather than downloading complete sources for post-hoc editing, each clip is specified prior to acquisition via three parameters: source URL, start timestamp, and duration. These parameters configure a pre-

cise temporal interval directly governing the retrieval process. Timestamp conversion into seconds is handled by `parse_time`, which reduces `HH:MM:SS` or `MM:SS` strings into integer values:

```
def parse_time(time_str):
    """Converts HH:MM:SS or MM:SS string to seconds."""
    parts = [int(p) for p in time_str.split(':')]

    if len(parts) == 3:
        return parts[0] * 3600 + parts[1] * 60 + parts[2]
    elif len(parts) == 2:
        return parts[0] * 60 + parts[1]
    return parts[0]
```

Adding the requested duration to the start position determines the upper boundary of the segment:

```
start_sec = parse_time(start_time_str)
end_sec = start_sec + int(durata_str)
```

The resulting bounds, `start_sec` and `end_sec`, define the exact temporal window extracted from the source. Output paths are structured prior to downloader configuration, ensuring MP4 container compliance and consistent dataset indexing under the target directory:

```
filename_input = input(
    "Enter output filename (e.g., clip): "
).strip()
if not filename_input.endswith(".mp4"):
    filename_input += ".mp4"
output = f"Dataset/video{filename_input}"
```

Prefixing filenames with `video` maintains a unified identifier convention across all dataset instances. Stream selection, temporal trimming, and media normalization are encapsulated within the `ydl_opts` dictionary. Format selection prioritizes high-resolution video streams

alongside optimal audio feeds:

```
ydl_opts = {
    'format': 'bestvideo[height<=2160]+bestaudio/best',
}
```

This filter selects video streams up to 2160p paired with the highest quality audio stream, falling back to pre-merged streams (`/best`) if separate tracks are unavailable. Temporal constraints are applied during acquisition via `download_range_func`:

```
    'download_ranges': download_range_func(
        None,
        [(start_sec, end_sec)]
    ),
```

Embedding segment bounds directly into the retrieval mechanism eliminates separate media cropping steps and reduces bandwidth overhead. To maintain frame-accurate cutting across keyframe-based compressed video formats, the pipeline enforces keyframe generation at boundary points:

```
    'force_keyframes_at_cuts': True,
```

Forcing keyframes ensures temporal alignment at cut points, preserving action continuity for downstream multimodal evaluation in MAIA-AV. Output formats and post-processing codecs are explicitly declared to standardize the corpus:

```
    'merge_output_format': 'mp4',
    'postprocessor_args': {
        'ffmpeg': [
            '-c:v', 'libx264',
            '-preset', 'fast',
            '-c:a', 'aac'
        ]
    },
```

FFmpeg re-encodes visual streams via H.264 (`libx264`, `fast` preset) and acoustic streams

via AAC, eliminating encoding heterogeneity across raw YouTube sources. The unified configuration dictionary combines acquisition parameters, formatting rules, output paths, and authentication credentials:

```
ydl_opts = {
    'format': 'bestvideo[height<=2160]+bestaudio/best',
    'download_ranges': download_range_func(
        None,
        [(start_sec, end_sec)]
    ),
    'force_keyframes_at_cuts': True,
    'merge_output_format': 'mp4',
    'postprocessor_args': {
        'ffmpeg': [
            '-c:v', 'libx264',
            '-preset', 'fast',
            '-c:a', 'aac'
        ]
    },
    'outtmpl': output,
    'cookiefile': 'src/youtube.com_cookies.txt',
}
```

Passing a session cookie file (`cookiefile`) enables retrieval of age-restricted or session-gated content without altering core extraction logic. Execution is wrapped in exception handling blocks, logging errors and returning non-zero exit codes upon failure:

```
try:
    with yt_dlp.YoutubeDL(ydl_opts) as ydl:
        ydl.download([url])
    print(f"\nSuccessfully downloaded clip to {output}")
except Exception as e:
    print(f"\nError: {str(e)}")
    sys.exit(1)
```

This setup allows `yt_dlp` to coordinate format resolution, stream extraction, interval clipping, and FFmpeg transcoding within a single execution pass.

Audio Conversion

In the MAIA-AV framework, the auditory element of each video is regarded as an autonomous source of information rather than a mere supplementary part of the visual narrative. Spoken language, ambient sounds, and acoustic phenomena offer significant evidence that enhances visual data and aids in scene interpretation. Isolating the audio allows this modality to be independently analyzed, ensuring its contribution is maintained throughout subsequent processing stages. For every video, the original audio track is extracted and archived as a separate file. This separation retains the identical temporal parameters and identifier associated with the corresponding audiovisual entity. Such a one-to-one mapping upholds alignment across modalities, enabling visual, acoustic, and textual representations to be traced back to the same foundational event. The resulting audio files function as direct inputs for further pipeline stages, including automatic speech transcription and specialized audio analyses. The extraction process neither generates a modified version of the original content nor alters it; instead, it separates the auditory channel, allowing independent processing while remaining completely synchronized with the originating video content.

Implementation Details The audio extraction stage converts each audiovisual dataset instance into a standalone acoustic representation while preserving multimodal alignment with the source video. Implemented via Python's `subprocess` module, the pipeline invokes FFmpeg to automate batch processing across the complete video collection. Input and output paths are declared explicitly at execution setup:

```
INPUT_FOLDER = Path("data/video")
OUTPUT_FOLDER = Path("data/audio")
VIDEO_EXTENSIONS = {".mp4"}
```

Decoupling input and output directories isolates raw video resources and establishes an aligned audio mirror directory. Restricting `VIDEO_EXTENSIONS` to MP4 matches the output format produced by the preceding acquisition stage. The core conversion routine is encapsulated within `mp4_to_mp3`:

```
def mp4_to_mp3(
    input_path: Union[str, Path],
    output_path: Optional[Union[str, Path]] = None,
    bitrate: str = "192k",
) -> Path:
    """Extract the audio from an MP4 file and save it as MP3."""
```

When an explicit destination path is omitted, the function retains the base filename and replaces the container extension:

```
    input_path = Path(input_path)
    output_path = (
        input_path.with_suffix(".mp3")
        if output_path is None
        else Path(output_path)
    )
```

This mapping strategy guarantees cross-modal alignment across representations (e.g., `video001.mp4` yields `video001.mp3`). Prior to invoking FFmpeg, parent directory existence is enforced:

```
    output_path.parent.mkdir(
        parents=True,
        exist_ok=True,
    )
```

This step allows nested input folder hierarchies to be automatically replicated within the output directory. Extraction is executed using a single `subprocess.run` call:

```
    subprocess.run(
        [
            "ffmpeg",
```

```
            "-y",
            "-i",
            str(input_path),
            "-vn",
            "-c:a",
            "libmp3lame",
            "-b:a",
            bitrate,
            str(output_path),
        ],
        check=True,
    )
```

The `-vn` flag discards video streams, ensuring that FFmpeg isolates the acoustic signal rather than performing generic media transcode operations. Acoustic streams are normalized using the `libmp3lame` codec at a constant target bitrate of 192 kbit/s (`-b:a 192k`). Passing `-y` overwrites pre-existing destination files without prompt interruption, facilitating seamless pipeline re-execution. Setting `check=True` ensures runtime errors raise a `CalledProcessError` rather than failing silently. Upon completion, the function returns the confirmed target path:

```
    return output_path
```

Batch processing begins by recursively indexing and sorting compatible media files:

```
def find_videos(folder: Path) -> list[Path]:
    """Return all supported video files found recursively."""
    return sorted(
        path
        for path in folder.rglob("*")
        if path.is_file()
        and path.suffix.lower() in VIDEO_EXTENSIONS
    )
```

Sorting file paths guarantees deterministic execution order across operating environments. Directory validity is verified before initializing conversion loops:

```
if not INPUT_FOLDER.exists():
    raise FileNotFoundError(
        f"Input folder not found: {INPUT_FOLDER}"
    )
videos = find_videos(INPUT_FOLDER)
if not videos:
    print(f"No videos found in: {INPUT_FOLDER}")
    return
```

Validating the input directory early avoids redundant FFmpeg invocations when processing empty or invalid paths. For each video, relative path resolution preserves source subdirectory structures under `OUTPUT_FOLDER`:

```
for index, video_path in enumerate(videos, start=1):
    relative_path = video_path.relative_to(INPUT_FOLDER)
    audio_path = (
        OUTPUT_FOLDER
        / relative_path.with_suffix(".mp3")
    )
```

Replicating source directory trees ensures structural parity between visual and acoustic datasets. Each file conversion is executed inside localized exception blocks:

```
    try:
        mp4_to_mp3(
            video_path,
            audio_path,
        )
        print("Audio extraction completed.")
    except subprocess.CalledProcessError as error:
        print(f"FFmpeg error: {error}")
```

Capturing `CalledProcessError` exceptions locally prevents single-file execution failures from interrupting broader batch operations while logging specific processing errors.

Transcription Extraction

Another key representation in our dataset is the audio transcription, following the methodology established in our previous work [4]. Transcriptions were generated using the *Faster-Whisper* model (`large-v3`), a reimplementation of *OpenAI's Whisper* powered by the *CTranslate2* fast inference engine for Transformer architectures. The model processes raw audio content directly, with the language parameter set to Italian. Additionally, a *Voice Activity Detection* (VAD) filter is applied to isolate segments containing actual speech, thereby minimizing non-speech transcriptions. The overall transcription procedure aims to convert spoken dialogue into textual content that can be reliably recognized and processed by downstream models. This methodology represents an evolution compared to previous approaches on acoustic content analysis [4], which jointly employed *Whisper-1* and *GPT-4o-transcribe* to compare and align outputs for higher precision. In contrast, the current pipeline streamlines the transcription workflow by deploying a single, locally executed model, providing an efficient and reproducible framework.

Implementation Details The transcription stage was implemented as an independent component of the data preparation pipeline to associate each video with a textual representation of its spoken content. Powered by `Faster-Whisper` and the `large-v3` model executed locally on GPU, this setup ensures full reproducibility while eliminating dependencies on external cloud services. The execution environment explicitly configures a selected CUDA device and establishes a local cache for model files:

```
    os.environ["CUDA_VISIBLE_DEVICES"] = "0"

    HF_CACHE = Path.cwd() / ".hf_cache"
    HF_CACHE.mkdir(parents=True, exist_ok=True)

    os.environ["HF_HOME"] = str(HF_CACHE)
```

```
    os.environ["HF_HUB_CACHE"] = str(HF_CACHE / "hub")
```

Utilizing a local cache confines model weights to the working directory, preventing redundant downloads across system locations. The transcription model is subsequently instantiated on CUDA using `float16` precision:

```
    MODEL_NAME = "large-v3"
    LANGUAGE = "it"

    model = WhisperModel(
        MODEL_NAME,
        device="cuda",
        device_index=0,
        compute_type="float16",
        download_root=str(HF_CACHE / "hub"),
    )
```

Setting `language="it"` explicitly conditions decoding on Italian. This parameter bypasses automatic language identification and directly aligns model execution with the linguistic profile of the dataset. Prior to processing, the pipeline recursively retrieves and filters all `.mp4` files within the input directory, sorting the resulting collection to guarantee deterministic execution order:

```
    def find_videos(folder: Path) -> list[Path]:
        return sorted(
            path
            for path in folder.rglob("*")
            if path.is_file()
            and path.suffix.lower() in VIDEO_EXTENSIONS
        )
```

The core transcription workflow relies on `model.transcribe()`. By directly parsing the audio embedded within the video container, Faster-Whisper eliminates intermediate extraction steps and yields chronologically ordered transcription segments:

```
    segments, _ = model.transcribe(
        str(video_path),
        language=LANGUAGE,
        beam_size=5,
        vad_filter=True,
        condition_on_previous_text=False,
    )
```

Three key hyperparameters govern decoding precision and stability:

- `beam_size=5`: Activates beam-search decoding, evaluating multiple competing hypotheses during generation to balance accuracy and computational overhead.
- `vad_filter=True`: Enables Voice Activity Detection to strip environmental noise, music, and unvoiced pauses, preventing non-linguistic signals from reaching the ASR engine.
- `condition_on_previous_text=False`: Disables cross-segment context conditioning, isolating individual speech segments and mitigating error propagation across the audio stream.

The resulting segment iterator is aggregated into a continuous transcript by stripping whitespace, discarding empty segments, and concatenating valid text chronologically:

```
    return " ".join(
        segment.text.strip()
        for segment in segments
        if segment.text.strip()
    )
```

Omitting downstream linguistic post-processing or semantic filtering preserves an unadulterated representation of raw ASR performance. The corpus is processed sequentially, using each video filename as a unique primary key:

```
    transcriptions = {}
```

```
    for index, video_path in enumerate(videos, start=1):
    try:
        transcription = transcribe_video(video_path)
        transcriptions[video_path.name] = transcription
    except Exception as error:
        transcriptions[video_path.name] = (
            f"ERROR: {error}"
        )
```

Robust error handling isolates processing failures on individual files, preventing execution halts and cleanly decoupling unvoiced media from runtime exceptions. Finally, extracted transcripts are exported to a structured CSV file indexed by `video_name` and `transcription`:

```
    writer = csv.DictWriter(
        csv_file,
        fieldnames=[
        "video_name",
        "transcription"
        ],
        delimiter=";",
    )
    writer.writeheader()
    writer.writerows(
        {
            "video_name": video_name,
            "transcription": transcription,
        }
        for video_name, transcription
        in transcriptions.items()
    )
```

2.2 Expanding the Questions and Answers Dataset

The construction of linguistic data in MAIA-AV mirrors the methodological framework established in the original MAIA benchmark. The process retains the question as the foundational element for subsequent answer collection and the creation of an evaluation dataset. In the original MAIA benchmark, each video was associated with 2,400 questions. Expert annotators, operating under specific category guidelines, were responsible for generating these questions. The questions were designed in an open-ended format to preclude simple binary responses, such as yes or no. Given that the original benchmark was intended for models based solely on visual input, annotators were explicitly instructed to disregard all acoustic data during the question formulation phase. MAIA-AV extends the original framework by modifying the informational basis for question generation. This version includes a total of 300 questions, with each video offering three inquiries that delve into spatial, temporal, and causal dimensions of the presented information. These categories are aligned with the structure of the original MAIA benchmark. The spatial category focuses on identifying locations or entities, either visible or referenced, along with detailing the spatial relationships between them. The temporal category examines event sequences, identifying when specific occurrences take place and establishing temporal relationships such as preceding, succeeding, or simultaneous actions. The causal category investigates the reasons behind or consequences of actions, events, or behaviors observed within the video context. The primary distinction between the original MAIA and its successor, MAIA-AV, is the integration of audio information, which is crucial in the latter version. Certain questions are specifically designed to address audio elements of the video, including character dialogue, vocal commands to infotainment systems, and ambient sounds. This addition incorporates linguistic content that connects information across both visual and auditory channels. The process of acquiring answers adheres to the MAIA guidelines. For the newly devised set of 300 questions, data was gathered through a questionnaire distributed among four volunteers. These individuals are native speakers of Italian, each born in Italy, and possessing at least a high school diploma. These participants engaged with 100 videos, responding to the corresponding 300 questions, thus

enabling the collection of independent responses and four distinct articulations of each answer. The structured approach remains intact: the videos operate as the multimodal information source, the questions identify the specific information to be extracted, and the human-generated answers provide the necessary linguistic reference for constructing subsequent components of the benchmark.

Caption-Foil Generation

Building upon the human-annotated responses, the construction of *MAIA-AV* proceeds with the automated generation of *caption–foil* pairs for the Visual Statement Verification (VSV) task. This procedure follows the methodology established in *MAIA*: each question-answer pair is first converted into a declarative true statement (*True Statement*, TS) and subsequently modified to create a minimally contrastive false statement (*False Statement*, FS). Consequently, the caption and foil share maximum linguistic structure, differing primarily in question-relevant information. While the original benchmark collects eight human responses per question to yield eight TS–FS pairs, *MAIA-AV* adopts a scaled-down approach: four independent responses per question yield four captions and four corresponding foils. With three questions per video (spatial, temporal, and causal), each video comprises twelve caption–foil pairs. The automated generation employs `GPT-4o-2024-08-06`, matching the exact model version used in the original *MAIA* pipeline. Generation proceeds in two distinct stages. In the initial stage, the model receives only the question and a single human answer. Rather than generating novel content, the prompt directs the model to synthesize the question–answer pair into a single declarative statement preserving the original semantic content. The instructions mandate minimal alterations to annotators' phrasing while requiring the conversion of first-person expressions into an impersonal stance. Accordingly, phrases like "I think that" or "I believe that" are removed from the caption without discarding any underlying uncertainty. Restricting outputs to concise declarative sentences leverages natural annotator variance while avoiding reliance on the model for factual core generation. This initial phase employs the prompt used for True Statement generation in *MAIA*:

```
Sei un assistente che crea affermazioni in italiano sui video.
```

```
Data una domanda Q e una risposta A relative a un video, devi creare
un'affermazione S basata su A.
Mentre generi S, cerca di non alterare le parole che compongono A.
Se A include verbi o frasi in prima persona (ad es., 'penso', 'credo'),
riformula S in modo impersonale, evitando la prima persona.
L'affermazione deve essere una frase breve e dichiarativa.[1]
```

The instructions include few-shot examples demonstrating question–answer fusion, with specific coverage of negative responses, non-deducible information, and expressions of uncertainty. Appended to the prompt, the target question and annotator response appear in `Q:` and `A:` format, conditioning the model to output solely the statement `S:`. The second stage processes the generated caption to yield the corresponding foil. Unlike the initial phase, this prompt adapts to the target question category. The core methodology relies on constructing *minimal pairs*: the model modifies only the information necessary to invalidate the caption, avoiding extraneous additions and preserving sentence structure. Consequently, distinguishing between caption and foil tests a model's ability to ground claims in audiovisual content rather than relying on surface linguistic cues. This design adheres to *MAIA*, directing GPT-4o to alter exclusively elements relevant to the specified semantic domain. Across all categories, the system prompt assigns a uniform role:

```
Sei un assistente progettato per creare foil a partire dalle caption.[2]
```

Subsequent instructions vary by targeted semantic dimension. For **spatial** questions, foil generation alters strictly the entity's position or location:

```
Data una caption in italiano (C) riguardante la posizione o la localizzazione
di qualcuno o qualcosa, il tuo compito è creare il suo foil (F) modificando
solo l'informazione spaziale.
```

[1]English translation: *You are an assistant that creates statements in Italian about videos. Given a question Q and an answer A concerning a video, you must create a statement S based on A. While generating S, try not to alter the words composing A. If A includes first-person verbs or phrases (e.g., 'I think,' 'I believe'), rephrase S to be impersonal, avoiding a first-person perspective. The statement should be a concise, declarative sentence.*

[2]English translation: *You are an assistant designed to create foils based on captions.*

`Non aggiungere altre informazioni rispetto a quanto dichiarato in C.`[3]

An accompanying example relocates a woman standing in a poppy field to a school setting. Preserving the subject and surrounding text isolates spatial information as the sole determinant of truth value. For **temporal** questions, instructions dictate altering the temporal relationships within the caption:

`Data una caption in italiano (C) riguardante gli eventi e quando avvengono, il tuo compito è creare il suo foil (F) modificando solo l'informazione temporale.`
`Non aggiungere altre informazioni rispetto a quanto dichiarato in C.`[4]

An illustrative example converts the chronological order of two events into an alternative sequence, preserving core semantic content while modifying event timing or order. For **causal** questions, foil construction alters the cause-and-effect relationship expressed in the caption:

`Data una caption in italiano (C) riguardante le cause o gli effetti degli eventi, il tuo compito è creare il suo foil (F) modificando le cause o l'effetto dell'evento principale.`
`Non aggiungere altre informazioni rispetto a quanto dichiarato in C.`[5]

The provided example retains the principal action, such as a person beginning to run, while altering solely the underlying cause. This focuses the contrast on the causal link

[3]English translation: *Given an Italian caption (C) regarding the position or location of someone or something, your task is to create its foil (F) by changing only the spatial information. Don't add other information with respect to what is stated in C.*

[4]English translation: *Given an Italian caption (C) regarding events and when they happen, your task is to create its foil (F) by changing only the temporal information. Don't add other information with respect to what is stated in C.*

[5]English translation: *Given an Italian caption (C) regarding the causes or the effects of events, your task is to create its foil (F) by changing the causes or the effect of the main event. Don't add other information with respect to what is stated in C.*

rather than arbitrary scene modifications. In the final task setup, option ordering is randomized, preventing positional bias and ensuring performance metrics reflect semantic comprehension rather than structural pattern exploitation.

Implementation Details The *caption–foil* generation pipeline is implemented in `caption_foil.py`. The script relies on `pandas` for tabular data processing and the OpenAI Python client for model inference. Both caption and foil generation utilize `gpt-4o-2024-08-06`, with authentication managed via the `OPENAI_API_KEY` environment variable. Input and output paths, along with the expected annotation structure, are defined at initialization:

```
MODEL = "gpt-4o-2024-08-06"
RAW_ANSWERS_DIR = Path("data/vsv/raw_human_answers")
QUESTIONS_FILE = Path("data/vsv/question_classification.csv")
OUTPUT_DIR = Path("data/vsv/caption-foil")
CHECKPOINT_FILE = OUTPUT_DIR / "caption_foil_checkpoint.csv"
OUTPUT_FILE = OUTPUT_DIR / "caption_foil.csv"
RANDOM_SEED = 42
EXPECTED_MACROAREAS = {"spaziale", "temporale", "causale"}
EXPECTED_ANNOTATORS = 4
client = OpenAI(api_key=os.environ["OPENAI_API_KEY"])
```

Questions are loaded from `data/vsv/question_classification.csv`. Only the video identifier, principal dimension, and question text are required for this stage:

```
questions = pd.read_csv(
    QUESTIONS_FILE,
    sep=";",
    encoding="utf-8-sig",
)
questions = questions[
    ["video_name", "principal_dimension", "question_text"]
].copy()
```

Prior to processing, the script verifies column availability, discards missing values, nor-

malizes category labels, and confirms the presence of only the three expected macroareas. Video identifiers are standardized using `normalize_video_name()`, which extracts the numeric identifier and converts it to the format used across the dataset:

```
def normalize_video_name(value):
    value = str(value).strip()
    match = re.search(r"(\d+)", value)
    if not match:
        raise ValueError(
            f"Impossibile ricavare il numero del video da: {value}"
        )

    return f"video{int(match.group(1)):03d}.mp4"
```

Additionally, the script ensures that each video is associated with exactly one spatial, one temporal, and one causal question. Duplicate video–macroarea pairs are rejected prior to generation. Human responses are automatically loaded from all CSV files within `data/vsv/raw_human_answers`. The file count is validated against the expected number of annotators:

```
files = sorted(RAW_ANSWERS_DIR.glob("*.csv"))
if len(files) != EXPECTED_ANNOTATORS:
    raise ValueError(
        f"Attesi {EXPECTED_ANNOTATORS} file di annotatori in "
        f"{RAW_ANSWERS_DIR}, trovati {len(files)}"
    )
```

Each file must contain the columns `titolo_video`, `macroarea`, and `risposta`. For backward compatibility, the alternative column name `risposte` is also accepted and automatically renamed. Annotator identifiers are extracted directly from the filenames:

```
if "risposta" not in df.columns and "risposte" in df.columns:
    df = df.rename(columns={"risposte": "risposta"})
df = df[["titolo_video", "macroarea", "risposta"]].copy()
```

```
df["video_name"] = df["titolo_video"].map(
    normalize_video_name
)
df["macroarea"] = (
    df["macroarea"]
    .astype(str)
    .str.lower()
    .str.strip()
)
df["annotator"] = file.stem.removeprefix("risposte_")
```

Questions and responses are subsequently aligned via the shared `video_name` and `macroarea` fields. The merge is constrained as *many-to-one*, matching multiple annotator responses to a single question:

```
data = answers.merge(
    questions,
    on=["video_name", "macroarea"],
    how="left",
    validate="many_to_one",
)
```

Additional consistency checks validate question keys against annotator responses. Missing answers or orphan responses interrupt execution prior to invoking any API requests. Responses corresponding to the same video and macroarea are grouped into a common pool. The pool identifier is formed by concatenating the video identifier with the corresponding macroarea:

```
data["video_id"] = (
    data["video_name"].str.removesuffix(".mp4")
)
data["pool_id"] = (
    data["video_id"] + "_" + data["macroarea"]
)
```

Each pool must contain exactly four responses. Individual entries are assigned a sequential position and a unique identifier:

```
data["pool_item"] = (
    data.groupby("pool_id").cumcount() + 1
)
data["id"] = (
    data["pool_id"]
    + "_"
    + data["pool_item"].astype(str).str.zfill(2)
)
```

API calls are centralized within `call_openai()`, which serves both caption and foil generation functions:

```
response = client.chat.completions.create(
    model=MODEL,
    messages=messages,
    max_tokens=max_tokens,
)
return response.choices[0].message.content.strip()
```

The function executes up to five attempts for recoverable API failures. Rate-limit errors trigger a 30-second sleep interval, whereas generic API errors pause execution for 10 seconds. Authentication errors terminate execution immediately:

```
except RateLimitError:
    if attempt == retries - 1:
        raise
    time.sleep(30)
except AuthenticationError:
    raise RuntimeError(
        "Errore di autenticazione: controlla OPENAI_API_KEY."
```

```
    )
except APIError:
    if attempt == retries - 1:
        raise
    time.sleep(10)
```

For each annotated response, caption and foil generation proceed sequentially. The former accepts the question and human answer as input, whereas the latter accepts the generated caption along with its associated macroarea:

```
caption = generate_caption(
    row["question_text"],
    row["risposta"],
)
foil = generate_foil(
    caption,
    row["macroarea"],
)
```

Outputs are post-processed via `clean_generation()`, which strips optional `S:` or `F:` prefixes returned by the model:

```
def clean_generation(text, prefix):
    text = text.strip()
    if text.startswith(f"{prefix}: "):
        text = text[3:]
    elif text.startswith(f"{prefix}:"):
        text = text[2:]

    return text.strip()
```

To accommodate long-running executions, the script implements a checkpointing mechanism. Cached entries are reused only if their question, annotator, and human answer match the current source data and both generated fields are non-null:

```
checkpoint_is_valid = (
    saved is not None
    and str(saved["question"]) == str(row["question_text"])
    and str(saved["annotator"]) == str(row["annotator"])
    and str(saved["human_answer"]) == str(row["risposta"])
    and pd.notna(saved["caption"])
    and pd.notna(saved["foil"])
)
```

Newly generated entries are appended incrementally to `caption_foil_checkpoint.csv`:

```
pd.DataFrame(generated_rows).to_csv(
    CHECKPOINT_FILE,
    index=False,
    encoding="utf-8-sig",
)
```

Following generation, caption and foil positions are randomized independently within each question category. A fixed random seed guarantees reproducibility while maintaining a balanced distribution of options across positions:

```
rng = random.Random(RANDOM_SEED)
for category, indices in (
    df.groupby("question_category").groups.items()
):
    indices = list(indices)
    n = len(indices)
    labels = [0] * (n // 2) + [1] * (n // 2)

    if n % 2:
        labels.append(rng.randint(0, 1))

    rng.shuffle(labels)
```

The resulting `target` variable indicates the position of the correct caption: a value of 0 places the caption in `answer1`, whereas a value of 1 assigns it to `answer2`:

```
df["answer1"] = df.apply(
    lambda row:
    row["caption"]
    if row["target"] == 0
    else row["foil"],
    axis=1,
)
df["answer2"] = df.apply(
    lambda row:
    row["foil"]
    if row["target"] == 0
    else row["caption"],
    axis=1,
)
```

The final dataset is saved to `data/vsv/caption-foil/caption_foil.csv`. Alongside the randomized options and target labels, the output retains all metadata required to trace each pair back to its originating video, question, annotator, and human response:

```
columns = [
    "id",
    "video_id",
    "video_name",
    "pool_id",
    "pool_item",
    "question_category",
    "question",
    "annotator",
    "human_answer",
    "caption",
    "foil",
```

```
    "answer1",
    "answer2",
    "target",
]
final_df[columns].to_csv(
    OUTPUT_FILE,
    index=False,
    encoding="utf-8-sig",
)
```

3. Study Design and Experimental Setup

3.1 Preliminary Analysis as a Diagnostic Tool for Multimodal Assessment

The preliminary analysis aims to make the MAIA assessment more diagnostic and less dependent on final task outcomes. Performance on a Visual Question Answering (VSV) item alone does not reveal which prerequisite capabilities required by the task have been successfully satisfied. A correct answer does not necessarily imply that the model has accurately identified the relevant entities, reconstructed the events, established the required temporal relations, or correctly integrated the available evidence. Conversely, an incorrect answer does not indicate which of these components is responsible for the failure. This limitation is particularly relevant for MAIA, which evaluates fine-grained reasoning grounded in video content. For this reason, the preliminary analysis is performed independently of the VSV task. Rather than relying exclusively on direct question answering and aggregated accuracy, the video content is first represented through a structured analysis of entities, events, and relations among them. This representation is subsequently compared with an analogous analysis of the requirements imposed by the questions. Separating the two stages reduces the risk that video analysis is conditioned by the linguistic formulation of the question or by the plausibility of the available answer options. The objective is therefore to determine **what information can be retrieved from the video and what level of integration is required to retrieve it**, before relating this evidence to the demands of the VSV task. This framework has a methodological affinity with *VILMA* [9], particularly with its distinction between preliminary proficiency evaluation and the main test. *VILMA* implements this principle through a caption–foil paradigm in a zero-shot setting, using proficiency tests to verify prerequisite capabilities before interpreting performance on more complex phenomena. Our approach does not reproduce the *VILMA* evaluation format; rather, it adopts the underlying methodological principle that complex

abilities should be interpreted in relation to the simpler perceptual, temporal, and semantic capabilities on which they depend. Accordingly, the proposed analysis distinguishes between tasks that require direct retrieval of localized information and tasks that require progressively greater integration across entities, events, temporal contexts, and inferential relations. In video understanding, this distinction is particularly important because relevant information may be contained within a single frame or distributed across a sequence. Questions belonging to the same nominal category may therefore impose substantially different processing requirements. A spatial question, for example, may require the retrieval of a location from a single frame or the reconstruction of a spatial configuration after a sequence of events. To capture this variation systematically, the classification is based on explicit operational criteria rather than on an intuitive notion of difficulty. The proposed question classification framework estimates the complexity of the operations required to retrieve and integrate the relevant evidence. A common three-level scale is applied to three dimensions: **spatial**, **temporal**, and **causal**. At Level 0, the required information can be retrieved directly without comparison or reconstruction. Level 1 requires the integration of an explicit relation or dependency involving a limited amount of evidence. Level 2 requires a broader reconstruction across events, temporal phases, state changes, or implicit relations. The same progression is applied consistently across the three dimensions, as summarized in Table 3.1. The classification criteria are grounded in properties of the evidence required to answer each question, including its temporal distribution, the number of relevant evidence units, the presence of event relations, and the explicit or implicit nature of the required information. These properties are intended to describe the informational requirements of the task rather than its linguistic formulation. An independent analysis of the video is therefore necessary to determine whether the entities, events, and relations presupposed by the question are actually recoverable from the available visual evidence. The video analysis is organized into successive stages of semantic representation. The preprocessing stage provides temporally indexed video content, including a dense representation used to preserve temporal information. Four omnimodal and multimodal models, subsequently evaluated on the VSV task, independently analyze temporally localized portions of each video. Their outputs are used to construct structured representa-

tions of the entities, events, and relations identified in the observed content. The analysis is performed independently of the questions so that the resulting representation reflects information retrieved from the video rather than evidence selected in response to a specific query. Temporally overlapping portions of the video are used to preserve the connection between extracted information and the intervals in which it is supported. This temporal anchoring is necessary for the subsequent comparison between the evidence represented by the models and the evidence required by individual questions. For each video, the analysis progressively considers three levels of representation: **entities**, **events**, and **relations between events**, including inferential relations where required. Outputs from the different models are initially kept separate, allowing inter-model agreement to be measured without prematurely merging their predictions. Agreement does not constitute ground truth, since different models may produce the same unsupported interpretation; it is instead treated as an indicator of **prediction stability**. The analysis therefore distinguishes information that is directly supported by observable evidence from information introduced through model inference. Within this framework, **event extraction** is treated as the identification of semantically structured events associated with specific temporal intervals. Because the same event may appear in multiple overlapping portions of the input, subsequent consolidation reduces redundancy while preserving genuinely repeated events. The consolidated representation can then be used to establish temporal relations such as *before*, *after*, *during*, and *overlap*, producing a temporally organized representation of the video. This distinction between entities and events is essential for evaluating temporal complexity. Knowing that a video contains a person, a mug, and a table provides a different type of information from knowing that the person picks up the mug, drinks from it, places it on the table, and subsequently walks away. The former represents the entities present in the scene, whereas the latter requires the representation of actions, temporal order, and changes of state. Similarly, identifying the location of an object at a given moment imposes different requirements from identifying its location after a particular event. The preliminary analysis is designed to make these differences explicit. The causal dimension introduces an additional inferential requirement. Temporal succession alone does not establish causality; consequently, causal relations must be distinguished from simple co-occurrence or temporal

ordering. The analysis first represents the relevant events and their temporal organization. It can then be determined whether answering a causal question requires information that is explicitly available in the video or an additional inference connecting observed events, states, or intentions. A further objective of the preliminary analysis is to determine which information can be recovered from the visual modality independently of other channels. This addresses a methodological issue highlighted by previous experiments [4], in which visual information was compared with additional modalities such as audio and automatic transcription. In the original MAIA setting, the addition of these channels did not produce a systematic improvement, while the inclusion of audio transcriptions resulted in a small but statistically robust performance decrease. One relevant characteristic of the original dataset is that its VSV items were constructed around visual content rather than around information specifically provided by the acoustic channel. For this reason, the preliminary analysis establishes a visual baseline focused on *what can be seen* before assessing the contribution of *what can be heard*. This separation makes it possible to evaluate whether acoustic information provides genuinely complementary evidence and whether questions designed for multimodal assessment require information distributed across more than one modality. The purpose of the preliminary analysis is therefore to characterize **which visual information is available in the video and how much evidence must be retrieved and integrated to answer a question**. This representation provides the basis for evaluating whether the models subsequently tested on the VSV task are able to recover and combine the information required by each item. In this regard, the preliminary analysis moves *MAIA-AV* beyond a purely result-based evaluation towards a more diagnostic and competence-oriented framework. Building on the methodological principle underlying the proficiency tests introduced by *VILMA*, the proposed approach makes the prerequisite conditions for successful question answering explicit through a structured analysis of video content. This allows model performance to be interpreted in relation to specific task requirements, distinguishing limitations in entity recognition, event identification, temporal integration, spatial reasoning, and causal inference rather than attributing an incorrect response generically to a failure of multimodal reasoning.

Category	Level 0	Level 1	Level 2
Spatial (S)	Direct retrieval of an entity's location or spatial property from a single frame.	Explicit spatial relation or event-conditioned localization involving at most two evidence units.	Spatial reconstruction across multiple phases, including movement, relocation, or state change.
Temporal (T)	Recognition of a single event or state without temporal comparison.	Temporal localization or pairwise ordering between two explicit events.	Multi-event temporal reconstruction, including duration, repetition, onset, or state evolution.
Causal (C)	Retrieval of an explicitly stated cause without inference.	Explicit cause–effect relation involving at most two evidence units.	Implicit or multi-event causal reconstruction requiring contextual or intentional inference.

Table 3.1: Complexity levels for spatial, temporal, and causal questions.

4. Results and Discussions

5. Conclusions

Conclusioni...

6. Future Perspectives

List of Abbreviations

VLM	Visual Language Model
VQA	Visual Question Answering
AGQA	Action Genome Question Answering
MVBench	Multi-modal Video Understanding Benchmark
VILMA	Video Language Model Assessment
MAIA	Multimodal AI Assessment
VSV	Visual Statement Verification
OEVQA	Open-Ended Visual Question Answering
MLM	Multimodal Language Model
AVQA	Audio-Visual Question Answering
Dave	Diagnostic benchmark for Audio Visual Evaluation
MMAU-Pro	Massive Multi-Task Audio Understanding and Reasoning benchmark, Pro version
SONIC-01	SOcial Natural Interaction Corpus (Omnimodal, v1)
GPT	Generative Pre-trained Transformer
MIRAGE	Assessing Hallucination in Multimodal Reasoning Chains of Multimodal Large Language Models

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Appendix

A. Appendix